

Buy the pattern, not the checkpoint.

MoPA-PD turns a privileged motion planner into a visual policy you can ship without it. Liu*, Uppal*, Sukhatme, Lim, Englert, Lee — CoRL 2021.

Push 100 / 32.0 / 110.8

Assembly 0 → 100

DR ≥ 96.7 sim-to-sim

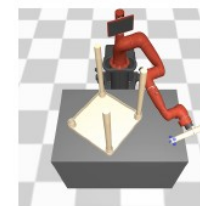
Sim has geometry. Deploy is vision. Clutter is the hard part. That is the buy. The 2021 Sawyer 32×32 checkpoint is not.



(a) Sawyer Push



(b) Sawyer Lift



(c) Sawyer Assembly

Five facts that do not change with the seat

Train privilege

01

Classical MP (RRT-Connect) lives inside MoPA-RL (Yamada et al., CoRL 2020). Geometry and the planner are train-time infrastructure, not the shipped controller.

Distill, then smooth

02

Visual BC on direct-action rollouts removes the planner and smooths RRT jitter. MoPA-PD then runs asymmetric SAC: actor from BC, critic from MoPA-RL Q , $Re:R\pi = 1:3$, small α .

Deploy is planner-free pixels

03

Inference is a 32×32 image plus joints. No RRT-Connect, no object state, no motion planner. The critic's state privilege ends when training ends.

Numbers you must be able to say

04

Push 100 / 32.0 / 110.8 (ASR / AEL / ADR). Assembly 0 without BC-smooth vs 100 with. $DR \geq 96.7$ sim-to-sim (Lift Scenario 1 is worst). Expert Push AEL 111 vs Ours 32.

What this paper is not

05

Not an ancestor of ACT, Diffusion Policy, π_0 , OpenVLA, or OXE — contrast poles only. Not a JEPA / LeWM / Cosmos child. Sim-only Sawyer. Do not launder DR into sim-to-real.

Classical robotics is still a training superpower

Obstructed exploration

Sparse contact around boxes and furniture is where pure visual SAC dies. Table 1: Asym. SAC and MoPA-Asym. SAC are 0. A planner that can see geometry is how the demos exist.

Productization, not the product

MoPA-RL already solves the tasks in state space. MoPA-PD's job is to throw the planner and the mesh away at deploy, and to shorten the path while doing it.

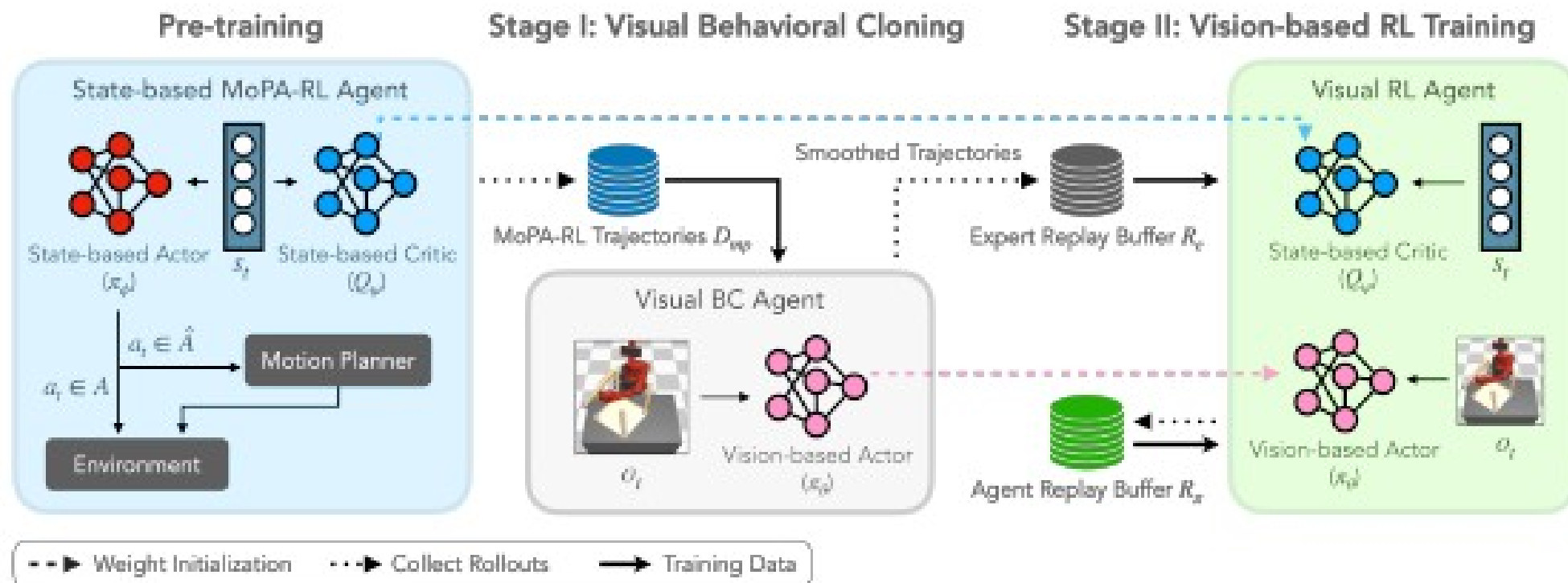
Not a TAMP stack

A distillation recipe for obstructed visuomotor control. Not a hierarchical planner library and not a BEHAVIOR skill API. Teach the pattern; do not assign the repo as the platform.

The alternative to demo-only

When teleop and OXE-scale data are thin and sim geometry is honest, privileged exploration beats waiting for more human demos. That is the Alternative-approaches argument.

Spend privilege in sim. Ship pixels.



Stage 0

MoPA-RL SAC. Small Δq is direct. Large Δq calls RRT-Connect. Full state.

Stage 1

Visual BC, MSE, 32×32. Drops the planner. Smoothed rollouts become the expert buffer R_e .

Stage 2

Asymmetric SAC. Actor from pixels. Critic from state. Init, low α , 1:3 mix.

Buy when geometry, vision deploy, and clutter line up

BUY THE PATTERN

Sim collision geometry is real

An RRT, TAMP, or MoPA-style teacher can run offline. If the mesh is a lie, Stage 0 dies.

Deploy must be vision

No planner latency and no object-state sensor on the robot. That is the product constraint.

Clutter is the bottleneck

Reach into a box, lift out, insert among obstacles. Pure visual RL does not explore this.

WALK TO ANOTHER POLE

Abundant teleop or OXE-scale data

Prefer ACT, Diffusion Policy, or a VLA. MoPA-PD is a sim recipe, not a replacement head.

No reliable collision model

Nothing privileged to distill. Do not invent a planner to justify the pipeline.

Cross-embodiment generality

One Sawyer in MuJoCo. Use the idea. Do not ship the checkpoint.

Same success. Much shorter paths than the expert.

	Sawyer Push			Sawyer Lift			Sawyer Assembly		
	ASR ↑	AEL ↓	ADR ↑	ASR ↑	AEL ↓	ADR ↑	ASR ↑	AEL ↓	ADR ↑
MoPA-RL	98.2	111.0	52.7	95.0	109.2	52.7	99.8	63.3	82.4
BC-Visual	99.4	118.0	46.9	62.0	108.8	34.6	97.0	115.1	50.4
Asym. SAC	0.0	250.0	0.0	0.0	250.0	0.2	0.0	250.0	3.4
MoPA-Asym. SAC	0.0	250.0	0.0	0.0	250.0	0.0	0.0	250.0	0.0
CoL	0.0	250.0	0.0	29.8	173.3	9.7	0.0	250.0	0.0
CoL w/ BC smoothing	0.0	250.0	0.0	0.0	250.0	5.1	0.0	250.0	0.0
Ours w/o BC smoothing	100.0	34.0	108.7	99.4	43.6	100.7	0.0	250.0	0.0
Ours	100.0	32.0	110.8	99.0	42.0	101.7	100.0	61.7	84.5

Table 1: Average success rate (ASR), episode length (AEL), and discounted return (ADR) of our method and baselines averaged over five seeds. Each method is evaluated after 3M environment steps. MoPA-RL [1] is the expert agent trained with state-based policy. All baselines below horizontal line are trained with asymmetric actor-critic for fair comparison. Maximum horizon is 250 for all tasks.

PUSH AEL

111 → 32.0

Expert MoPA-RL vs MoPA-PD. Push ADR moves 52.7 → 110.8. The policy is not cloning the planner. It is shorter.

ASR is the wrong headline on Push and Assembly: BC-Visual is already 99.4 and 97.0. The buy is AEL / ADR, plus Lift (BC 62 → 99).

Four costs the abstract does not waive

Sim-only. Tax unpaid.

The conclusion names real-world fine-tune and a vision critic as future work. Table 2 is sim-to-sim. Do not brief this as a robot result.

32×32, one arm

Sawyer 7-DoF, MuJoCo, three tasks, tiny RGB. Not wrist cameras, not multi-embodiment, not a household camera stack.

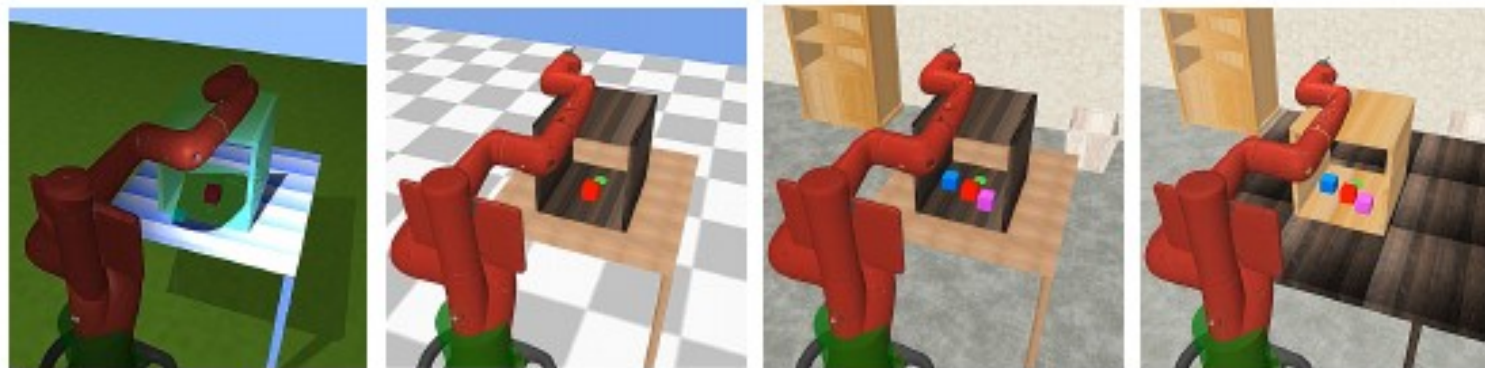
Stage 0 is engineering

You need a working MoPA-RL or another MP teacher, plus collision checks. Frozen demos are the practical path. The planner is not free.

The critic is still privileged

Asymmetric actor-critic (Pinto et al.) sees full state while learning. “Vision-only” is an inference claim, not a training claim.

Robust in sim. Not a sim-to-real receipt.



(a) Domain randomization

(b) Original environment

(c) Scenario 1

(d) Scenario 2

	Sawyer Push			Sawyer Lift			Sawyer Assembly		
	ASR ↑	AEL ↓	ADR ↑	ASR ↑	AEL ↓	ADR ↑	ASR ↑	AEL ↓	ADR ↑
Original env	99.7	39.1	103.1	99.3	37.3	106.5	100.0	49.7	93.4
Scenario 1	100.0	40.4	102.2	96.7	38.3	103	100.0	47.9	94.8
Scenario 2	99.3	43.4	99.04	97.3	37.0	104.7	100.0	47.7	95.0

Table 2: Evaluation metrics for domain transfer to unseen environments, illustrated in Figure 4.

DO NOT LAUNDER

Worst cell is Lift, Scenario 1: ASR 96.7.
The paper's claim is >96% on three
unseen sim scenarios. Appearance shift
is not a real robot.

Where modern stacks win, say so

ACT / ALOHA

Chunked CVAE cloning from human teleop. Different data, different action interface. Use it when demonstrations are what you actually have.

Diffusion Policy

Generative action chunks from demos. Not a privileged-planner distill. Table 1 does not beat it — that baseline did not exist here.

π_0 / OpenVLA / OXE

Large multi-task vision-language-action. MoPA-PD is single-task sim distillation. No cited ancestry. Do not brief one.

DART, separately

DART labels recoveries by disturbing the expert. MoPA-PD amortizes a planner. Shared covariate-shift vocabulary. Not the same product.

Pattern in. Checkpoint out.

In

An MP or TAMP teacher when sim geometry is trustworthy. BC smoothing before any RL buffer. A vision actor at deploy. Path length, not just success.

Out

2021 32×32 weights. Online RRT on the robot. A claim that DR solved reality. Any sentence that starts “this is why VLAs work.”

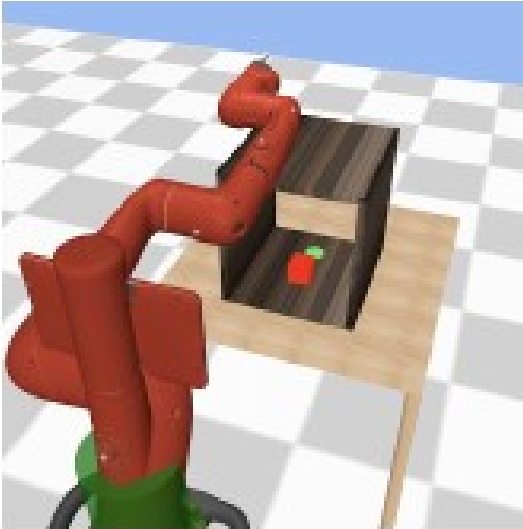
Budget shape

Paper protocol: 3M steps × 5 seeds × 3 tasks, after a 1M-step teacher. The only printed times are BC (Table 8): 30 / 39 / 120 min. Full RL GPU time is unstated.

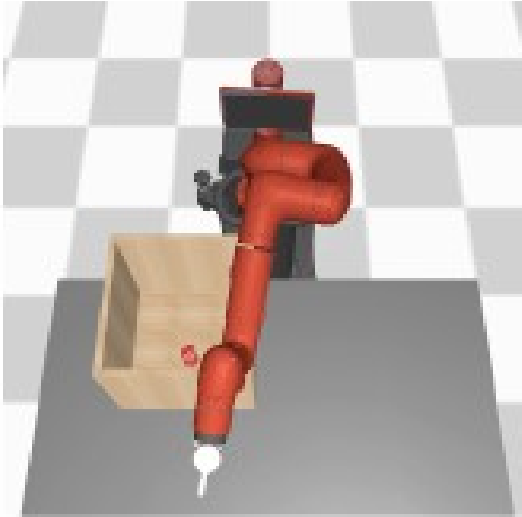
Decision

Fund the pattern on the next cluttered sim task that already has a mesh. Do not fund a Sawyer reproduction as a product milestone.

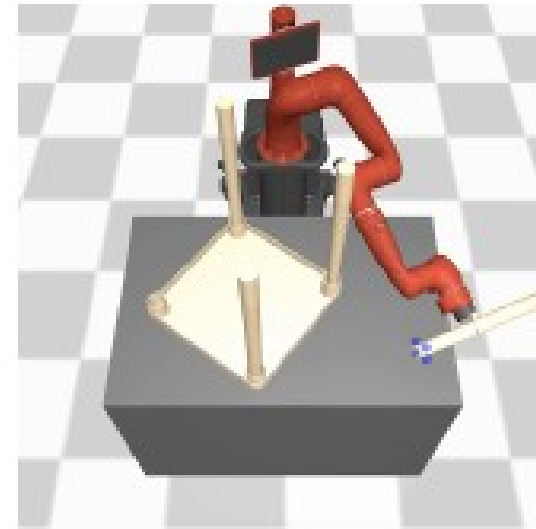
Three obstructed Sawyer tasks. Nothing else was scored.



(a) Sawyer Push



(b) Sawyer Lift



(c) Sawyer Assembly

Push

Reach into a box. Push the red cube onto the green goal. ASR is easy for BC. Path length is not.

Lift

Reach, grasp, lift a can out without collisions. BC-Visual ASR falls to 62. Stage 2 earns its place.

Assembly

Insert the fourth table leg among the other three. IKEA furniture env. Dies without BC smoothing.

Sentences that survive contact with the PDF

Say

“Train-time RRT-Connect inside MoPA-RL, distilled to a visual policy.”

Say

“Push episodes went from 111 steps to 32, at full success.”

Say

“Domain randomization held up in unseen sim scenes, worst cell 96.7.”

Don't say

“We have a sim-to-real manipulation system.”

Don't say

“This is an early VLA / ACT / diffusion policy.”

Don't say

“It works on BEHAVIOR.” Nothing in the paper ran that benchmark.

Say this, then stop

ONE-LINER

Lock the privileged-planner → vision-student alternative before the pure VLA weeks. Buy the pattern. Do not buy the 2021 Sawyer checkpoint.

Remember

Push 100 / 32.0 / 110.8. Assembly dies at 0 without BC smoothing.

Refuse

Sim-to-sim DR is not sim-to-real. No invented robot success rate.